

Notes: Antràs, Fort & Tintelnot (AER, 2017)

The Margins of Global Sourcing: Theory and Evidence from US Firms

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data construction and measurement	4
3.1	Main firm-level data sources	4
3.2	Import products, source countries, and sourcing sets	4
3.3	Input purchases and domestic sourcing	4
3.4	Country characteristics and external data	5
3.5	Outcome groups for China-shock analysis	5
4	Model and empirical specifications	5
4.1	Preferences and demand	5
4.2	Input sourcing and sourcing capability	6
4.3	Marginal cost and firm profit	6
4.4	Complementarity condition	6
4.5	Sourcing potential regression	7
4.6	Trade elasticity regression	7
4.7	Simulated method of moments for fixed costs	7
4.8	China-shock reduced-form specification	7
5	Identification and empirical strategy	8
5.1	Structural identification logic	8
5.2	Counterfactual identification logic	8
5.3	Reduced-form validation strategy	8
5.4	Threats and how the paper addresses them	9
6	Tables and figures	9
6.1	Figure 1: Sales Premia and Minimum Number of Sourcing Countries in 2007; Table 1: Top Ten Source Countries for US Firms, by Number of Firms	9
6.2	Table 2: Firm-Level Statistics on the Number of Source Countries per Imported Product; Table 3: US Firms Importing from Strings of Top Ten Countries	10
6.3	Figure 2: Country Sourcing Potential Parameters and the Extensive Margin; Table 4: Estimation of Firm and Aggregate Trade Elasticities	10
6.4	Table 5: Estimated Parameters; Figure 3: Estimated Sourcing Potential and Median Fixed Costs by Country	11
6.5	Figure 4: Model Fit - Share of Firms and Aggregate Sourcing by Country; Table 6: Hierarchies in Sourcing Patterns - Data and Model	12

6.6	Table 7: Third Country Sourcing Effects of Chinese Sourcing Potential Shock; Figure 5: Changes in the Size of Firms	13
6.7	Table 8: Gross and Net US Sourcing Effects; Table 9: Observed Changes in Third Country Sourcing from 1997-2007 by Firms' 2007 Chinese Import Status	13
6.8	Table 10: Estimates of the Impact of the China Shock on Firm-Level Sourcing	14
7	Interpretive synthesis	15
8	Short summary	15
9	Conclusion	16
9.1	Core conclusion	16
9.2	Economic intuition	16
9.3	Incremental contribution	16
9.4	What the paper does not claim	16
9.5	Final takeaway	17

1 Big picture

- **Question.** How do heterogeneous firms choose which foreign countries to source inputs from, and how do these extensive-margin sourcing choices shape the response to trade shocks?
- **Core problem.** Importing inputs differs from exporting final goods. In canonical export models with constant marginal costs, the decision to sell in one destination can be analyzed market-by-market. In global sourcing, each additional source country changes the firm’s marginal cost, so the gain from one source country depends on the other source countries already in the firm’s sourcing set.
- **Main theoretical insight.** Under the empirically relevant case in which the dispersion of input efficiencies is high relative to demand curvature, sourcing decisions are *complements*. A firm that adds one source country becomes more likely to add others, because marginal cost reductions raise output scale and make further sourcing investments more valuable.
- **Main empirical setting.** The paper uses confidential US Census Bureau data linking manufacturing firms’ production data to import transactions by country and HS10 product. This allows the authors to observe which firms import, from which countries, and how much they import.
- **Main structural contribution.** The model converts a potentially impossible 2^J discrete-choice problem into a tractable problem by exploiting complementarity and adapting Jia’s (2008) iterative algorithm to a multi-country sourcing environment.
- **Main quantitative result.** The estimated model implies that sourcing from all foreign countries lowers a firm’s variable costs by about 9 percent and raises sales by about 32 percent relative to sourcing only domestically.
- **Main counterfactual.** A China sourcing shock calibrated to match the growth in the Chinese import share generates both substitution away from US inputs and firm-level expansion among China importers. This creates gross domestic sourcing increases for some firms even when aggregate US sourcing falls.
- **Reduced-form validation.** Firms receiving larger exogenous China sourcing shocks increase not only Chinese imports but also domestic and third-country sourcing, which is difficult to reconcile with independent country-by-country sourcing decisions.
- **Economic takeaway.** Global sourcing shocks are not just import-substitution shocks. Because sourcing sets are complementary, a shock to one country can reorganize a firm’s entire sourcing network and generate heterogeneous effects across firms.

2 Incremental contribution of the paper

- **From exporting to sourcing.** The paper shifts the extensive-margin literature from export destinations to import source countries. This matters because sourcing directly changes marginal cost, so choices interact through the production function.
- **Multi-country extensive margin.** Prior global sourcing models often focused on two countries or partial characterizations. This paper provides a quantifiable framework with many countries, many inputs, heterogeneous firms, and country-specific fixed costs.
- **Complementarity as a solution device.** The paper shows that when $\theta < \sigma - 1$, extensive-margin sourcing choices have complementarities. This result is not just economic intuition; it is what makes the high-dimensional discrete-choice problem computationally feasible.
- **Country sourcing potential versus fixed cost.** The paper separates two country-level forces that are often conflated: a country’s variable-cost advantage as a source of inputs and the fixed cost of accessing that country. Canada, China, Mexico, Germany, and Japan differ along these two dimensions in ways that explain why importer counts and import values rank countries differently.

- **Structural and reduced-form bridge.** The paper does not rely only on a structural model. It compares the model’s China-shock predictions to observed firm-level sourcing changes from 1997 to 2007 and to a reduced-form IV strategy inspired by the China-shock literature.
- **New view of domestic effects of offshoring.** A key novelty is that importing from China can raise a firm’s domestic sourcing if the productivity/scale effect dominates substitution. This yields a more nuanced view than simply treating imports as replacing domestic inputs one-for-one.
- **Methodological portability.** The algorithmic approach can be used in other economic settings with complementary discrete choices, such as exporting with nonconstant marginal costs, plant location, supplier networks, or demand-linked entry decisions.

3 Data construction and measurement

3.1 Main firm-level data sources

- The main micro data come from the US Census Bureau’s Economic Censuses, the Longitudinal Business Database, and the Import Transaction Database.
- The LBD provides employment and industry information for private, nonfarm US establishments.
- The Economic Census provides establishment-level variables such as sales, value added, and input usage.
- The import transaction data cover the universe of US import transactions and identify the importing firm, country of origin, product category, and value.
- The analysis focuses on firms with at least one manufacturing establishment, because the model is about goods production using intermediate inputs.

Interpretation: The data are unusually powerful because they connect a firm’s production outcomes to its detailed sourcing portfolio. The paper can observe not just whether a firm imports, but the countries and products that make up its sourcing strategy.

3.2 Import products, source countries, and sourcing sets

- Products are measured at the HS10 level; the import data contain nearly 17,000 product categories.
- A source country is a foreign country from which a firm imports at least one input.
- A sourcing strategy $\mathcal{J}_i(\varphi)$ is the set of countries from which firm φ pays the fixed cost to source.
- In 2007, importers source an average of about 12 products from about three foreign countries; the median importer sources two products and two countries; the 95th percentile importer sources 41 products and 11 countries.
- The model assumes each input is sourced from the lowest-cost location. Table 2 supports this assumption: the median firm imports a given HS10 product from only one country.

Interpretation: The product-level evidence supports an Eaton-Kortum style selection process: a firm may use many countries across different inputs, but each individual input is usually sourced from one country.

3.3 Input purchases and domestic sourcing

- Total input purchases combine production-worker wages from the Census of Manufactures, total material costs from Economic Censuses, and merchandise purchases from the Census of Wholesale.

- Foreign input purchases from country j are measured as imports from country j .
- Domestic input purchases equal total input purchases minus total imports.
- Firm-level input shares are constructed as foreign or domestic purchases divided by total input purchases.
- For reduced-form comparisons over time, nominal values are deflated using NBER industry deflators where possible.

Interpretation: Domestic sourcing is not directly observed as a separate supplier-level variable. It is constructed as a residual from total input purchases and imported inputs, so measurement quality depends on accurately matching firm production and import data.

3.4 Country characteristics and external data

- Country characteristics include wages, distance, R&D, capital per worker, common language, population, control of corruption, and number of firms.
- Wages are human-capital adjusted using schooling and returns-to-schooling information.
- Population is used as an instrument for wages in the trade-elasticity estimation.
- Corruption and institutional measures help capture fixed-cost and variable-cost barriers to sourcing.
- HS-product-level trade flows are mapped to NAICS industries using concordances such as Pierce and Schott.

Interpretation: These variables allow the paper to distinguish two forces: countries may be good sources because they are low-cost or technologically attractive, but they may also be costly to access because of distance, language, institutions, or firm-country-specific fixed costs.

3.5 Outcome groups for China-shock analysis

- **Entrants** are firms that begin sourcing from China after the shock.
- **Continuers** are firms that source from China before and after the shock.
- **Others** are firms that do not source from China before or after the shock.
- For observed data comparisons, the paper uses firm sourcing changes from 1997 to 2007, spanning China's WTO accession and the rapid increase in US sourcing from China.

Interpretation: This grouping is central because the model predicts sharply different domestic and third-market responses depending on whether the firm enters, continues, or remains outside China sourcing.

4 Model and empirical specifications

4.1 Preferences and demand

Consumers have CES preferences over manufacturing varieties:

$$U_i^M = \left(\int_{\omega \in \Omega_i} q_i(\omega)^{(\sigma-1)/\sigma} d\omega \right)^{\sigma/(\sigma-1)}, \quad \sigma > 1. \quad (1)$$

Demand for a variety is

$$q_i(\omega) = E_i P_i^{\sigma-1} p_i(\omega)^{-\sigma}. \quad (2)$$

Interpretation: The elasticity of demand σ determines how strongly firm sales and profits react to marginal-cost reductions from better sourcing.

4.2 Input sourcing and sourcing capability

The cost of procuring input v from the lowest-cost country in the firm's sourcing set is

$$z_i(v, \varphi; \mathcal{J}_i) = \min_{j \in \mathcal{J}_i} \{\tau_{ij} a_j(v, \varphi) w_j\}. \quad (3)$$

With Frechet input efficiencies, the share of intermediate purchases sourced from country j is

$$\chi_{ij}(\varphi) = \frac{T_j(\tau_{ij} w_j)^{-\theta}}{\Theta_i(\varphi)} \quad \text{if } j \in \mathcal{J}_i(\varphi), \quad (4)$$

where

$$\Theta_i(\varphi) = \sum_{k \in \mathcal{J}_i(\varphi)} T_k(\tau_{ik} w_k)^{-\theta}. \quad (5)$$

Interpretation: Θ_i is the firm's sourcing capability. Adding a country increases Θ_i , lowers marginal cost, and changes the profitability of sourcing from all other countries.

4.3 Marginal cost and firm profit

The firm's marginal cost can be written as

$$c_i(\varphi) = \frac{1}{\varphi} (\gamma \Theta_i(\varphi))^{-1/\theta}. \quad (6)$$

The source-set problem is to choose countries to maximize operating profits net of fixed costs:

$$\max_{\mathcal{J} \subseteq J} \left\{ B_i \left(\varphi (\gamma \Theta_i(\mathcal{J}))^{1/\theta} \right)^{\sigma-1} - w_i \sum_{j \in \mathcal{J}} f_{ij} \right\}. \quad (7)$$

Interpretation: The firm faces a combinatorial problem because every subset of countries implies a different sourcing capability and a different fixed-cost burden. With many countries, direct search over all 2^J possible subsets is infeasible.

4.4 Complementarity condition

The key complementarity case is

$$\theta < \sigma - 1. \quad (8)$$

Under this condition, the profit gain from adding one source country is larger when the firm already sources from more countries.

Interpretation: When demand is elastic and input efficiency differences are sufficiently dispersed, lower marginal costs translate strongly into scale expansion. This makes other countries more attractive, so sourcing choices become complements rather than substitutes.

4.5 Sourcing potential regression

The country sourcing-potential parameters are recovered from firm-country input shares. In simplified form, the estimating relationship maps firm import shares to country fixed effects and firm observables:

$$\log \chi_{ij} = \text{firm term} + \log \xi_j + \text{measurement error}_{ij}. \quad (9)$$

Interpretation: The country fixed effects identify average sourcing potential: countries with high fixed effects are attractive sources of marginal-cost reduction, conditional on firms selecting into those countries.

4.6 Trade elasticity regression

The paper estimates a country-level gravity-style regression:

$$\log \hat{\xi}_j = \beta_0 + \beta_r \log R\&D_j + \beta_k \log \text{capital}_j + \beta_F \log \text{firms}_j - \theta \log w_j - \theta \beta_d \log \text{distance}_j + \beta_C \text{corruption}_j + \iota_j. \quad (10)$$

Wages are instrumented with country population in the preferred IV specification.

Interpretation: The wage coefficient identifies the firm-level trade elasticity θ . Instrumenting wages with population addresses the concern that high wages may reflect unobserved country productivity or infrastructure.

4.7 Simulated method of moments for fixed costs

The model estimates fixed-cost parameters by simulated method of moments. The moments target:

- the overall share of firms that import;
- the share of importers from each country;
- the share of imports by country;
- the sourcing hierarchy statistics.

Interpretation: The moments force the model to fit both extensive-margin and intensive-margin import patterns. This is what allows the model to distinguish country sourcing potential from fixed costs.

4.8 China-shock reduced-form specification

The paper estimates

$$\Delta y_n = \beta_0 + \beta_{Ch} \Delta \text{China}_n + \varepsilon_n, \quad (11)$$

where ΔChina_n is a Davis-Haltiwanger-Schuh growth rate in firm n 's Chinese imports from 1997 to 2007.

The instrument for firm-level Chinese import growth is based on changes in Chinese export shares to EU15 countries in the firm's 1997 input industries.

Interpretation: This reduced-form design asks whether exogenous improvements in a firm's Chinese sourcing opportunities are associated with expansion or contraction in domestic and third-market sourcing.

5 Identification and empirical strategy

5.1 Structural identification logic

1. **Descriptive restrictions.** The paper first establishes facts that support the model: firms source many inputs, typically source each product from one country, and display hierarchy in source-country choice.
2. **Sourcing potential.** Conditional on observed sourcing sets, firm-country import shares identify country sourcing potential because the share formula maps relative purchases to relative variable costs.
3. **Trade elasticity.** Cross-country variation in wages and sourcing potential identifies the elasticity θ , with population used as an instrument for wages.
4. **Fixed costs.** The mismatch between country importer counts and import values identifies fixed costs: a country can have large import values but relatively few importers if it has high sourcing potential and high fixed costs.
5. **Firm-country heterogeneity.** The model requires idiosyncratic fixed-cost variation because common country-level fixed costs cannot fit observed importer shares, import shares, and hierarchy patterns simultaneously.

Interpretation: The identification strategy uses different parts of the data for different parameters. Import shares reveal variable-cost attractiveness; importer counts and hierarchy patterns reveal fixed costs and heterogeneity.

5.2 Counterfactual identification logic

- The China sourcing shock is calibrated to match the observed 178 percent increase in the Chinese import share from 1997 to 2007.
- The calibrated structural model then predicts firm-level changes in domestic sourcing, third-country sourcing, imports from China, sales, and aggregate sourcing.
- Alternative models are recalibrated to match the same China import-share growth, so differences in predictions reflect model structure rather than simply shock size.
- The key comparison is whether the model generates gross increases in domestic and third-country sourcing for China importers while also producing aggregate domestic decline.

Interpretation: The counterfactual is disciplined by an observed aggregate fact, but its value comes from the cross-firm predictions: who grows, who shrinks, and where gross reallocations occur.

5.3 Reduced-form validation strategy

- The reduced-form exercise uses changes in Chinese exports to EU15 countries as a source of variation in Chinese sourcing potential by input industry.
- This shock is mapped to firms based on their 1997 input industry structure, before the main 1997-2007 sourcing changes are realized.
- Outcomes include changes in domestic input purchases, foreign inputs from countries other than China, the number of source countries, and firm employment.
- Standard errors are clustered at the industry level, matching the level at which the China shock varies.

Interpretation: If Chinese sourcing merely substitutes for domestic inputs, the coefficient on domestic sourcing should be negative. The paper instead finds positive effects on domestic and third-market sourcing, supporting the complementarity mechanism.

5.4 Threats and how the paper addresses them

- **Demand or productivity shocks.** Firms that expand Chinese imports may also be growing for unobserved reasons. The IV strategy uses external changes in Chinese export shares to EU15 markets to reduce this concern.
- **Output-market competition.** Chinese export growth to Europe may proxy for Chinese competition in final goods. The paper examines robustness to controlling for Chinese import penetration and alternative instruments.
- **Input-supplier shocks.** The instrument may affect suppliers in third countries indirectly. The paper acknowledges this limitation; richer supplier-network data would help separate these channels.
- **Structural assumptions.** The model assumes Frechet technology, fixed costs, and wages pinned down outside manufacturing. Sensitivity analysis varies key parameters and shows qualitative robustness.
- **Measurement of domestic inputs.** Domestic sourcing is residualized from total inputs and imports, so measurement error is possible. The qualitative pattern is strengthened by comparing structural predictions with reduced-form evidence.

Interpretation: The paper combines structural discipline with reduced-form validation. Neither component alone is perfect, but the agreement between the two strengthens the argument that extensive-margin interdependencies are economically relevant.

6 Tables and figures

6.1 Figure 1: Sales Premia and Minimum Number of Sourcing Countries in 2007; Table 1: Top Ten Source Countries for US Firms, by Number of Firms

Read:

- Figure 1 shows a strong monotone relationship between firm sales premia and the minimum number of source countries.
- Table 1 shows that the rank of countries by number of importers differs sharply from the rank by value of imports.
- Canada is first by both firms and value, but China is second by firm count and third by value; Mexico is eighth by firm count and second by value.

Interpretation: Figure 1 motivates a model with selection: larger and more productive firms source from more countries. Table 1 motivates the need for two country-level dimensions. A country can be attractive on the intensive margin but costly to enter on the extensive margin.

Economic message: The data reject a one-dimensional view of countries. Sourcing potential and fixed entry costs both matter.

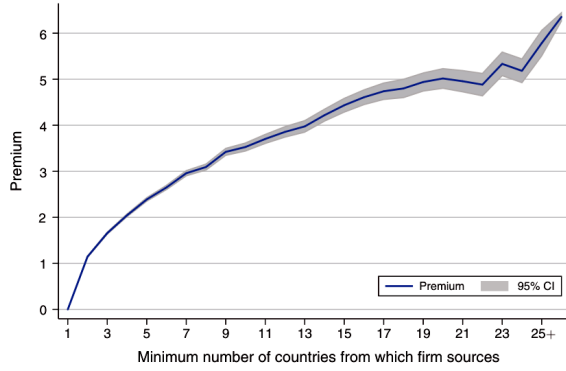


FIGURE 1. SALES PREMIA AND MINIMUM NUMBER OF SOURCING COUNTRIES IN 2007

(a) Figure 1: Sales premia and sourcing countries

TABLE 1—TOP TEN SOURCE COUNTRIES FOR US FIRMS, BY NUMBER OF FIRMS

	Rank by		Number of importers		Value of imports	
	Firms	Value	Firms	Percent of total	Imports	Percent of total
Canada	1	1	37,810	59	145,740	16
China	2	3	21,460	33	121,990	13
Germany	3	5	13,050	20	62,930	7
United Kingdom	4	6	11,450	18	30,750	3
Taiwan	5	11	10,490	16	16,630	2
Italy	6	13	8,500	13	13,230	1
Japan	7	4	7,980	12	112,250	12
Mexico	8	2	7,810	12	125,980	14
France	9	9	6,110	9	22,980	3
Korea, South	10	10	5,610	9	20,390	2

Notes: Sample is US firms with some manufacturing activity in 2007. Number of firms rounded to nearest 10 for disclosure avoidance. Imports in millions of USD, rounded to nearest 10 million for disclosure avoidance.

(b) Table 1: Top source countries by importer counts and values

6.2 Table 2: Firm-Level Statistics on the Number of Source Countries per Imported Product; Table 3: US Firms Importing from Strings of Top Ten Countries

Read:

- Table 2 shows that firms usually import a given HS10 product from a single country.
- Table 3 shows that 36 percent of importers follow the top-country pecking order, compared with 19.6 percent under random entry.
- The pecking-order pattern is present but far from universal, pointing to both common country attractiveness and firm-specific idiosyncrasies.

Interpretation: Table 2 supports the lowest-cost-supplier logic of the model. Table 3 supports the complementarity and hierarchy predictions but also shows why firm-country-specific fixed costs are needed.

Economic message: The sourcing model must fit both hierarchy and heterogeneity; a purely common ordering of countries would be too rigid.

TABLE 2—FIRM-LEVEL STATISTICS ON THE NUMBER OF SOURCE COUNTRIES PER IMPORTED PRODUCT

	Firm level		
	Mean	Median	Max
Mean	1.11	1.03	1.78
Median	1.00	1.00	1.00
95th percentile	1.61	1.00	4.00

Note: Table reports statistics on the number of countries from which a firm imports the same HS10 product.

(a) Table 2: Source countries per imported product

TABLE 3—US FIRMS IMPORTING FROM STRINGS OF TOP TEN COUNTRIES

String	Data		Random entry	
	Firms	Percent of importers	Firms	Percent of importers
CA	17,980	29.82	6,760	11.21
CA-CH	2,210	3.67	3,730	6.19
CA-CH-DE	340	0.56	1,030	1.71
CA-CH-DE-GB	150	0.25	240	0.40
CA-CH-DE-GB-TW	80	0.13	50	0.08
CA-CH-DE-GB-TW-IT	30	0.05	10	0.02
CA-CH-DE-GB-TW-IT-JP	30	0.05	0	0.00
CA-CH-DE-GB-TW-IT-JP-MX	50	0.08	0	0.00
CA-CH-DE-GB-TW-IT-JP-MX-FR	160	0.27	0	0.00
CA-CH-DE-GB-TW-IT-JP-MX-FR-KR	650	1.08	0	0.00
TOTAL following pecking order	21,680	36.0	11,820	19.6

Notes: The string CA means importing from Canada but no other among the top ten; CA-CH means importing from Canada and China but no other; and so forth. Percent of importers shows percent of each category relative to all firms that import from ten ten countries

(b) Table 3: Sourcing hierarchy among top ten countries

6.3 Figure 2: Country Sourcing Potential Parameters and the Extensive Margin; Table 4: Estimation of Firm and Aggregate Trade Elasticities

Read:

- Figure 2 shows that estimated sourcing potential is strongly related to the number of firms importing from a country.
- China has high sourcing potential, consistent with its broad appeal to US firms.
- Table 4 estimates the trade elasticity using firm-level and aggregate variation, with IV specifications using population as an instrument for wages.
- The preferred firm-level IV estimate of θ is 1.789.

Interpretation: Figure 2 provides validation for the estimated source-country attractiveness parameters. Table 4 supplies a key structural parameter determining whether sourcing choices are complements or substitutes.

Economic message: The low value of θ relative to $\sigma - 1$ places the model in the complementarity region.

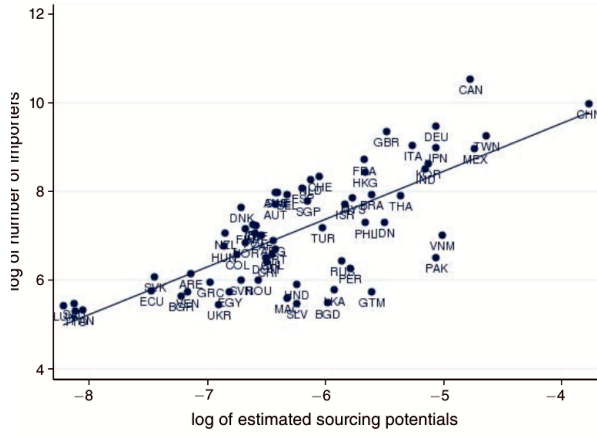


FIGURE 2. COUNTRY SOURCING POTENTIAL PARAMETERS AND THE EXTENSIVE MARGIN

(a) Figure 2: Sourcing potential and extensive margin

	log ξ		log aggregate imports		
	OLS	IV	OLS	IV	IV
log HC adjusted wage	-0.537 (0.184)	-1.789 (0.696)	-0.643 (0.390)	-4.544 (1.844)	-1.268 (0.768)
log distance	-0.341 (0.197)	-0.621 (0.294)	-0.859 (0.418)	-1.733 (0.779)	-0.650 (0.333)
log R&D	0.352 (0.068)	0.524 (0.125)	0.763 (0.144)	1.298 (0.332)	0.251 (0.176)
log capital/worker	-0.184 (0.175)	0.425 (0.390)	-0.264 (0.370)	1.633 (1.033)	0.308 (0.421)
Common language	0.105 (0.223)	0.146 (0.289)	0.354 (0.471)	0.479 (0.764)	0.137 (0.317)
Control corrupt	0.156 (0.151)	0.621 (0.312)	0.365 (0.319)	1.816 (0.826)	0.414 (0.350)
log number of firms	0.108 (0.086)	-0.020 (0.130)	0.031 (0.183)	-0.369 (0.345)	-0.062 (0.142)
log domestic purchases				2.392 (0.327)	
Constant	-7.250 (0.922)	-11.068 (2.323)	14.499 (1.952)	2.600 (6.156)	-37.389 (6.573)
Observations	57	57	57	57	57

Notes: Standard errors in parentheses. In the IV specifications, the human-capital-adjusted wage is instrumented by population. HC adjusted wage is country wage adjusted for differences in human capital. Domestic purchases is total purchases of US inputs by firms sourcing from a country. First-stage F -statistic on the excluded instrument is

(b) Table 4: Firm and aggregate trade elasticities

6.4 Table 5: Estimated Parameters; Figure 3: Estimated Sourcing Potential and Median Fixed Costs by Country

Read:

- Table 5 reports simulated method of moments estimates for fixed-cost determinants.
- Fixed costs rise with distance and fall with common language.
- Figure 3 plots sourcing potential against median fixed costs, showing that countries differ along both dimensions.
- Canada has high sourcing potential and low fixed costs, while China has high sourcing potential but higher fixed costs.

Interpretation: The model fits Table 1 because it separates countries that many firms can access cheaply from countries that are attractive but costly. Firm-country idiosyncratic fixed costs explain why not all firms follow the same sourcing hierarchy.

Economic message: The extensive margin is shaped by access costs as much as by variable-cost advantages.

TABLE 5—ESTIMATED PARAMETERS

B	β_c^f	β_d^f	β_l^f	β_c^f	β_{disp}^f
0.122 (0.004)	0.022 (0.002)	0.193 (0.018)	0.872 (0.024)	-0.393 (0.012)	0.934 (0.018)

Notes: Table reports coefficients and standard errors from estimating the model via simulated method of moments. Standard errors based on 25 bootstrap samples drawn with replacement.

(a) Table 5: Estimated fixed-cost parameters

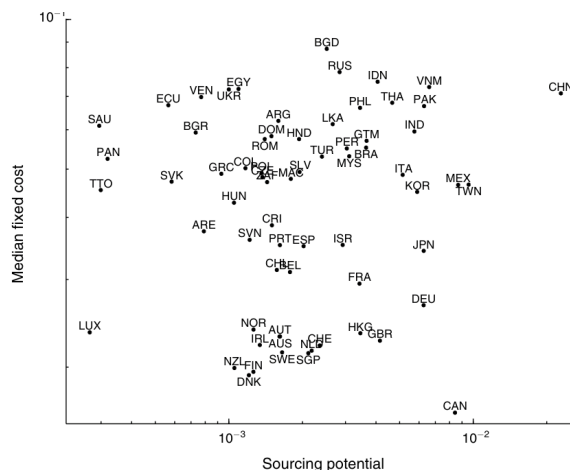


FIGURE 3. ESTIMATED SOURCING POTENTIAL AND MEDIAN FIXED COSTS BY COUNTRY

(b) Figure 3: Sourcing potential and fixed costs

6.5 Figure 4: Model Fit - Share of Firms and Aggregate Sourcing by Country; Table 6: Hierarchies in Sourcing Patterns - Data and Model

Read:

- Figure 4 shows that the model fits the share of importers by country very well and also fits aggregate foreign sourcing shares reasonably well.
- Table 6 shows that the baseline model approximates the observed hierarchy of sourcing patterns.
- The model does better than a common-fixed-cost alternative because it allows country and firm-country heterogeneity.

Interpretation: The model is not only matching aggregate import shares. It also captures which firms source from which countries, which is essential for counterfactuals that depend on the extensive margin.

Economic message: Good fit on extensive-margin moments is what makes the model useful for analyzing firm-level sourcing shocks.

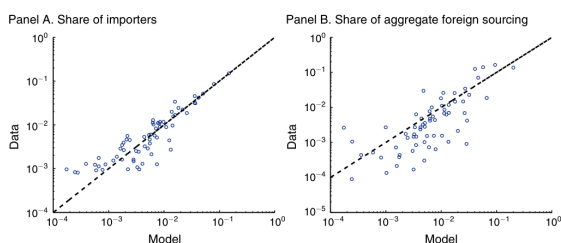


FIGURE 4. MODEL FIT: SHARE OF FIRMS AND AGGREGATE SOURCING BY COUNTRY

(a) Figure 4: Model fit

TABLE 6—HIERARCHIES IN SOURCING PATTERNS: DATA AND MODEL

String	Data	Baseline model
CA	29.82	28.66
CA-CH	3.67	3.42
CA-CH-DE	0.56	0.58
CA-CH-DE-GB	0.25	0.14
CA-CH-DE-GB-TW	0.13	0.09
CA-CH-DE-GB-TW-IT	0.05	0.02
CA-CH-DE-GB-TW-IT-JP	0.05	0.03
CA-CH-DE-GB-TW-IT-JP-MX	0.08	0.07
CA-CH-DE-GB-TW-IT-JP-MX-FR	0.27	0.12
CA-CH-DE-GB-TW-IT-JP-MX-FR-KR	1.08	0.66
TOTAL following pecking order	36.0	33.8

Notes: This table depicts the percentage of importers following a particular sourcing pattern. The first row shows the percentage of firms only importing from Canada; the second row shows the percentage of firms only importing from Canada and China; and so forth (irrespective of firm sourcing outside these top ten countries). The ranking of countries is determined by the number of firms sourcing from these countries in the data.

(b) Table 6: Hierarchy fit in data and model

6.6 Table 7: Third Country Sourcing Effects of Chinese Sourcing Potential Shock; Figure 5: Changes in the Size of Firms

Read:

- Table 7 shows that new China importers increase sourcing from the United States and from other foreign countries in the baseline model.
- In independent-entry or universal-importing models, this gross expansion is absent or much smaller.
- Figure 5 shows that the China shock increases sales mostly for large firms and shrinks sales for much of the lower part of the size distribution.

Interpretation: The China shock reallocates activity across firms. Large or productive firms that can profitably add China expand and increase other sourcing; firms that cannot benefit contract because they face tougher competition.

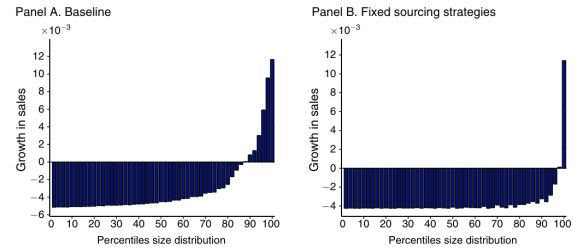
Economic message: Import shocks create within-industry reallocation, not just aggregate substitution away from domestic production.

TABLE 7—THIRD COUNTRY SOURCING EFFECTS OF CHINESE SOURCING POTENTIAL SHOCK

Chinese import status	Change in sourcing from US	Change in sourcing from other countries	Change in sourcing from China	Share of firms	Share of imports from China
<i>Panel A. Baseline model</i>					
Entrants	1.008	1.015	∞	0.053	0.334
Continuers	1.001	1.001	2.148	0.027	0.666
Others	0.995	0.987	—	0.920	0.000
<i>Panel B. Baseline model, fixed sourcing strategies</i>					
Entrants	—	—	—	0.000	0.000
Continuers	1.002	1.002	2.150	0.027	1.000
Others	0.996	0.996	—	0.973	0.000
<i>Panel C. Independent entry decisions model ($\theta = \sigma - 1 = 2.85$)</i>					
Entrants	0.997	0.994	∞	0.053	0.343
Continuers	0.997	0.996	2.116	0.027	0.657
Others	0.997	0.992	—	0.920	0.000
<i>Panel D. Universal importing model (no fixed costs)</i>					
Entrants	—	—	—	0.000	0.000
Continuers	0.987	0.987	2.976	1.000	1.000
Others	—	—	—	0.000	0.000
<i>Panel E. Common fixed costs across countries model (heterogeneous across firms)</i>					
Entrants	1.006	—	∞	0.108	0.328
Continuers	0.999	0.999	2.249	0.066	0.672
Others	0.993	—	—	0.826	0.000

Notes: The table groups firms by Chinese import status. Entrants are those firms (i.e., bundles of productivity levels and fixed cost draws) that begin sourcing from China. Continuers are firms that source from China before and after the shock. Others are firms that do not source from China before or after the shock. As described in the main text, the shock to the Chinese sourcing potential is calibrated so that the baseline model matches the observed 178 percent increase in the Chinese import share from 1997 to 2007. For the panels C, D, and E, we re-calibrate the shock to the Chinese sourcing potential so that the respective models can match the same 178 percent increase in the Chinese import share. Columns 1, 2, and 3 contain the ratio of the total sourcing by each group of firms before and after the shock. Figures in this table do not include changes in sourcing due to changes in the mass of firms after the counterfactual shock. Changes in sourcing due to changes in the overall mass of firms are included

(a) Table 7: Third-country sourcing effects of China shock



(b) Figure 5: Changes in firm size distribution

6.7 Table 8: Gross and Net US Sourcing Effects; Table 9: Observed Changes in Third Country Sourcing from 1997-2007 by Firms' 2007 Chinese Import Status

Read:

- Table 8 shows that aggregate US sourcing falls by 0.53 percent, but some firms increase domestic sourcing by US\$4.41 billion.
- The gross increase in domestic sourcing offsets part of the gross decrease by contracting or exiting firms.
- Table 9 shows that the same qualitative pattern is visible in actual data: entrants into China sourcing grow domestic and other-country sourcing more than other firms.

Interpretation: Table 8 highlights why net aggregate effects are misleading: they hide large gross flows in opposite directions. Table 9 validates the model by showing that the pattern predicted by the structural counterfactual appears in the observed 1997-2007 data.

Economic message: The domestic effect of importing from China is heterogeneous. Some domestic input suppliers lose demand, while others gain demand from expanding China importers.

TABLE 8—GROSS AND NET US SOURCING EFFECTS

	Baseline		Fixed sourcing strategies	
	Difference in sourcing in billion USD	Change in percent of total US sourcing	Difference in sourcing in billion USD	Change in percent of total US sourcing
Increase in domestic sourcing	4.41	0.11	1.97	0.05
Decrease in domestic sourcing by firms that continue to operate	-13.93	-0.36	-12.86	-0.33
Decrease in domestic sourcing by firms that shut down	-11.11	-0.28	0	0
Total	-20.63	-0.53	-10.88	-0.28

Notes: We use the model's predictions for the US sourcing pre and post the shock to China's sourcing potential in order to calculate percentage changes in US sourcing. We then use aggregate US sourcing purchases in the data and the percentage differences predicted by the model to calculate the predicted USD change in sourcing in response to the shock.

(a) Table 8: Gross and net US sourcing effects

TABLE 9—OBSERVED CHANGES IN THIRD COUNTRY SOURCING FROM 1997–2007 BY FIRMS' 2007 CHINESE IMPORT STATUS

Chinese import status	Change in sourcing from United States	Change in sourcing from other countries	Change in sourcing from China	Share of firms	Share of imports from China
Entrants	2.68	5.93	—	0.08	0.37
Exiters	0.15	0.02	0.00	0.00	0.00
Continuers	1.03	1.32	5.75	0.01	0.63
Others	0.74	1.03	—	0.91	0.00

Notes: This table is based on an unbalanced panel of manufacturing firms from 1997 and 2007. Exiters (entrants) are those firms that stop (begin) sourcing from China. Continuers are firms that source from China in 1997 and 2007. Columns 1, 2, and 3 contain the ratio of total sourcing in 2007 deflated dollars relative to 1997 dollars by each group of firms. The share of firms in column 4 is the number of 2007 firms in each China import status category over total active firms in 2007.

(b) Table 9: Observed sourcing changes by China-import status

6.8 Table 10: Estimates of the Impact of the China Shock on Firm-Level Sourcing

Read:

- Table 10 reports OLS and IV estimates of the relationship between firm-level China sourcing growth and other sourcing outcomes.
- The IV estimates imply that China sourcing growth increases domestic inputs, the number of source countries, and foreign inputs from countries other than China.
- The IV employment effect is negative but statistically insignificant, which the paper interprets cautiously because the model's outcome is total input sourcing rather than firm-level employment itself.

Interpretation: The reduced-form evidence rules out the simplest substitution story. Increased China sourcing is associated with broader sourcing expansion, consistent with complementary extensive-margin sourcing choices.

Economic message: The empirical China shock evidence supports the structural mechanism: sourcing from one country can raise sourcing from others.

TABLE 10—ESTIMATES OF THE IMPACT OF THE CHINA SHOCK ON FIRM-LEVEL SOURCING

Firm-level change from 1997 to 2007 in								
	OLS				IV			
	Domestic inputs	Number of countries	Foreign inputs	Firm empl.	Domestic inputs	Number of countries	Foreign inputs	Firm empl.
China, DHS	0.064 (0.010)	0.255 (0.007)	0.362 (0.013)	0.097 (0.007)	0.758 (0.214)	0.551 (0.080)	0.670 (0.198)	−0.092 (0.162)
Constant	0.054 (0.019)	0.144 (0.013)	0.315 (0.026)	−0.075 (0.014)	−0.054 (0.039)	0.098 (0.017)	0.267 (0.044)	−0.046 (0.032)
Observations	127,400	127,400	127,400	127,400	127,400	127,400	127,400	127,400
First-stage statistics	Coefficient (se) 2.685 (0.505)				KP F -stat 28.25			

Notes: All variables are changes or growth rates from 1997 to 2007. China, DHS is a Davis-Haltiwanger-Schuh growth rate in firm imports from China. Domestic inputs, foreign inputs, and firm employment are a DHS growth rate. Number of countries is the log difference in the number of countries (excluding China, but including the United States) from which the firm sources inputs. Foreign inputs exclude China. Standard errors are in parentheses and clustered by 439 NAICS industries. In the IV specifications, firm-level sourcing from China is instrumented by the change in Chinese market share in EU15 countries of a weighted average of the firm's inputs. KP F -stat is the Kleibergen Paap F -statistic. The number of observations is rounded for disclosure avoidance.

Table 10: Reduced-form estimates of the China shock

7 Interpretive synthesis

- **Complementarity changes the meaning of offshoring.** If sourcing choices were independent, a China shock would mainly substitute away from domestic or third-country suppliers. With complementarity, firms that add China may also expand other sourcing because lower costs raise firm scale.
- **Fixed costs produce heterogeneous treatment effects.** The same China shock affects firms differently depending on productivity and preexisting sourcing sets. Incumbent and entrant importers benefit; firms outside China sourcing face competitive pressure.
- **Extensive margins matter quantitatively.** Much of the response comes from firms changing which countries they source from, not only changing the value of imports conditional on a fixed sourcing set.
- **Gross flows matter more than net aggregates.** Aggregate domestic sourcing falls, but that net decline hides sizable domestic sourcing increases by expanding importers.
- **Data and theory discipline each other.** Descriptive facts motivate the model; structural estimates quantify it; reduced-form evidence checks whether the central mechanism is visible in actual changes.

8 Short summary

- The paper builds a multi-country global sourcing model with heterogeneous firms, multiple inputs, variable trade costs, and country-specific fixed costs.
- Firms choose sourcing sets, not just sourcing volumes.
- The central challenge is that with J countries there are 2^J possible sourcing sets.
- Under $\theta < \sigma - 1$, sourcing choices are complements, allowing the authors to solve the high-dimensional firm problem.
- The data show that US importers source many products from a few countries, usually one country per product.

- Country ranks by number of importers and import value differ sharply, requiring separate sourcing-potential and fixed-cost components.
- The model estimates country sourcing potential, trade elasticity, and fixed-cost parameters using confidential Census data.
- Estimated fixed costs are higher with distance and lower with common language.
- The model fits importer shares, aggregate sourcing shares, and sourcing hierarchies reasonably well.
- A China shock calibrated to the observed 1997-2007 growth of Chinese imports generates heterogeneous firm responses.
- New and continuing China importers expand domestic and third-country sourcing, while non-China firms shrink.
- Reduced-form evidence using a firm-level China sourcing shock supports the same qualitative mechanism.
- The paper’s main conclusion is that extensive-margin sourcing interdependencies are economically important for understanding trade shocks.

9 Conclusion

9.1 Core conclusion

Antràs, Fort, and Tintelnot show that firms’ global sourcing decisions are interdependent across countries. Because importing inputs lowers marginal cost, sourcing from one country can raise the value of sourcing from other countries. This makes the extensive margin of importing fundamentally different from the extensive margin of exporting in canonical constant-marginal-cost models.

9.2 Economic intuition

The intuition is a competition between two forces. A new source country can substitute for existing suppliers, but it can also lower costs, expand the firm, and make additional sourcing relationships worthwhile. In the estimated model, the expansion and complementarity force is important enough that China importers can increase domestic sourcing even while aggregate domestic sourcing falls.

9.3 Incremental contribution

The paper’s incremental contribution is to make this mechanism tractable, estimable, and quantitatively relevant. It provides the first quantitative multi-country characterization of firm extensive-margin sourcing with multiple inputs and heterogeneous fixed costs; it separates sourcing potential from fixed access costs; and it validates the mechanism using both structural counterfactuals and reduced-form evidence.

9.4 What the paper does not claim

The paper does not claim that offshoring always raises domestic sourcing or employment. Aggregate domestic sourcing falls in the China counterfactual. The key claim is more subtle: the net decline masks large heterogeneous gross responses, with expanding China importers increasing domestic and third-market purchases while non-importers contract.

9.5 Final takeaway

The main lesson is that trade shocks propagate through firms' sourcing networks. Understanding global production requires modeling the extensive margin of input sourcing as a network of complementary choices, not as a collection of independent country-by-country import decisions.